

Maguette Sembene

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EDUCATION

Virginia Tech, Department of Agricultural and Applied Economics <i>Ph.D. in Applied Economics</i>	Expected 2027
Virginia Tech, Department of Agricultural and Applied Economics <i>Master of Science in Agricultural and Applied Economics</i>	2023
École Nationale Supérieure d'Agriculture de Thiès, Senegal <i>Agricultural Engineering Degree</i>	2020

FIELDS

Development Economics, Environmental and Climate Economics, Applied Econometrics

RESEARCH EXPERIENCE

Virginia Tech Graduate Research Assistant	2021 – Present
- Built and managed a three-wave household panel database (1,100+ households, 4,600+ GPS-measured plots across 75 villages): harmonized survey instruments across waves, integrated secondary geospatial and climate data, and maintained reproducible data-cleaning pipelines in R and Stata - Designed and executed econometric analyses (panel methods, instrumental variables) linking climate and remote-sensing data to agricultural and household outcomes; published results in <i>Food Policy</i> - Produced data visualizations, policy briefs, and reports for government and donor audiences, including a sector review used by regulators and seed-industry stakeholders - Conducted primary data collection with producers, traders, and regulators; drafted IRB protocols and informed-consent documents for large-scale household surveys	
Feed the Future Innovation Lab for Peanut (USAID) Undergraduate Research Assistant	2020 – 2021
- Collected primary household survey data through in-person interviews with groundnut farmers in Senegal's Groundnut Basin - Analyzed survey data using R statistical software to identify gender and age disparities in groundnut production revenues - Produced 50-page undergraduate thesis analyzing patterns in agricultural gender disparities with policy implications	

PUBLICATIONS AND RESEARCH PAPERS

Published

- Sembene, M., Mills, B., & Gupta, A. (2025). Economic costs of extreme heat on groundnut production in the Senegal Groundnut Basin. *Food Policy*, 136, 102970.

Working Papers

- “Short-Term Changes in Land Access in the Senegal Groundnut Basin.” With B. Mills and A. Gupta.
- “Land Access, Climate Variability, and Youth Exit from Farming in the Senegal Groundnut Basin.” (**Job Market Paper**). With B. Mills and A. Gupta.
- “Seed Quality, Information Failure, and Groundnut Productivity in Senegal.” With B. Mills and A. Gupta.

Work in Progress

- “Did Cheap Digital Payments Reach Rural Senegal? Price, Distance, and Mobile Money Access”

Other Research and Policy Briefs

- “Groundnut Exports from Senegal to China: Domestic Effects.” Sembène, M. (2025). Research paper.
- “Senegal Groundnut Seed Sector Review” with B. Mills (2024). Report.
- “Gestion de la matière organique du sol dans le bassin arachidier du Sénégal.” Mbaye, T., Mills, B., Touré, K., Sembene, M., Diatta, P., & Diédhiou, I. (2022). Policy brief.
- “Utilisation des médias par les jeunes et perceptions du climat et des pratiques agricoles.” Sembène, M., Mbaye, T., Touré, K., & Mills, B. (2023). Policy brief.

TECHNICAL SKILLS

Programming & Statistical Software: R (tidyverse, fixest, sf), Python (pandas, geopandas), Stata, MATLAB

Data & Visualization Tools: Tableau, QGIS, LaTeX

Languages: English and French (fluent), Wolof (native)

CONFERENCES AND PRESENTATIONS

“Short-Term Changes in Land Access in the Senegal Groundnut Basin” 2025

- Agricultural and Applied Economics Association (AAEA) Annual Meeting, Denver, CO

“Economic Costs of Extreme Heat on Groundnut Production in the Senegal Groundnut Basin” 2023

- Department Seminar, Virginia Tech Department of Agricultural and Applied Economics

PROFESSIONAL SERVICE

Peer Reviewer

- *Food Policy*
- *Agricultural Finance Review*

TEACHING EXPERIENCE

Virginia Tech, Department of Agricultural and Applied Economics

- Teaching Assistant, *Fundamentals of Econometrics* (AAEC/STAT 4804, AAEC 5804G) – Spring 2026
- Teaching Assistant, *Applied Economic Forecasting* (AAEC 4484, AAEC/STAT 5484) – Spring 2026

GRANTS AND FELLOWSHIPS

Paul J. Driscoll Fellowship 2023, 2024

Graduate Assistantship at Virginia Tech 2021 – Present

Senegalese Government Excellence Scholarship 2016 – 2021

HONORS AND AWARDS

Outstanding Master’s Thesis Award, VT Graduate School 2023

Award for Innovation at the University of Thiès, Senegal 2020

Winner of the Student Scientific Article Writing Competition 2018

ADDITIONAL PROFESSIONAL EXPERIENCE

Ferme Wa Keur Khadim | *Research Intern* 2019

- Conducted land prospection and socio-economic analysis; performed parcel mapping using GPS technology for spatial analysis

Regional Inspection of Water and Forests of Dakar | *Research Intern* 2018

- Identified coordination mechanisms and stakeholders within regional environmental services

École Nationale Supérieure d’Agriculture de Thiès | *Research Intern* 2016

- Analyzed community sociology and economics; participated in agricultural fieldwork and reported findings

REFERENCES

Dr. Bradford Mills

Advisor, Professor

Department of Agricultural and Applied Economics
Virginia Tech

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Dr. Anubhab Gupta

Assistant Professor

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